

QCE Chemistry Units 3&4

Test 2: Chemical equilibrium systems

- Brønsted–Lowry model
- Dissociation constants
- Acid–base indicators
- Volumetric analysis

SECTION 1 – MULTIPLE CHOICE QUESTIONS

QUESTION 1 C

C is correct. A Brønsted–Lowry acid is a proton donor. In this equation, HCl donates a proton to H_2O , forming H_3O^+ .

A and **B** are incorrect. In these equations, the species in bold accept a proton, meaning they are acting as a Brønsted–Lowry base.

D is incorrect. This equation is not an acid–base reaction.

QUESTION 2 A

A is correct. A Brønsted–Lowry base accepts protons. In this reaction, water (H_2O) accepts a proton, becoming H_3O^+ .

B, **C** and **D** are incorrect. These reactions show H_2O donating a proton, meaning that it is acting as a Brønsted–Lowry acid.

QUESTION 3 C

C is correct. Propanoic acid is a carboxylic acid; it does not fully dissociate in water so it is weak.

A and **B** are incorrect. Hydrochloric acid and sulfuric acid are strong acids; they fully dissociate in water.

D is incorrect. Ammonia solution is a weak base.

QUESTION 4 B

B is correct. An amphoteric substance can accept and donate protons. HCO_3^- (hydrocarbonate ion) can accept a proton to become H_2CO_3 or donate a proton to become CO_3^{2-} .

A is incorrect. HNO_3 cannot accept a proton and be stable.

C is incorrect. NaOH cannot donate a proton; the hydroxides in alkali metals cannot donate protons.

D is incorrect. NaCl is a salt and does not contain a proton.

QUESTION 5 C

C is correct. A conjugate acid is the substance formed by a base accepting a proton. Ammonia is the original base and becomes an ammonium ion after accepting a proton from hydrochloric acid. Therefore, the ammonium ion is the conjugate acid.

A and **B** are incorrect. Hydrochloric acid and ammonia are the reactants; they are the original acid and base, respectively.

D is incorrect. The chloride ion is what remains after the hydrochloric acid donates a proton to ammonia, so it is the conjugate base.

QUESTION 6 A

A is correct. Sulfuric acid (H_2SO_4) is diprotic; therefore, even though its concentration is 0.3 M, its hydrogen ion (H^+) concentration would be 0.6 M.

B is incorrect. Ethanoic acid (CH_3COOH) is a weak acid and only partially dissociates, leading to a very low H^+ concentration.

C is incorrect. Nitric acid (HNO_3) is monoprotic, which means it can donate only one H^+ per molecule; therefore, while its concentration (0.4 M) is higher than 0.3 M H_2SO_4 , its H^+ concentration is lower.

D is incorrect. Sodium hydroxide (NaOH) is a base and would, therefore, have a very low H^+ concentration.

QUESTION 7 C

C is correct. H_2O has a H^+ concentration of 1×10^{-7} M; only a very small number of H_2O molecules self-ionise.

A and **B** are incorrect. H_2SO_4 and HCl are strong acids and would have high H^+ concentrations due to full dissociation.

D is incorrect. CH_3COOH is a weak acid, but it would dissociate more and still have a much higher H^+ concentration than H_2O .

QUESTION 8 B

B is correct. Bromophenol blue appears yellow at pH values less than 3.

A and **D** are incorrect. Bromophenol blue is never red or colourless.

C is incorrect. Bromophenol blue appears blue at pH values greater than 4.6.

QUESTION 9 A

A is correct and **B** is incorrect. There is a 1 : 1 ratio of the two reactants; however, the concentration of the acid (ethanoic acid) is double the concentration of the base (sodium hydroxide). Therefore, the student would require 10 mL of ethanoic acid, which is half the volume of the sodium hydroxide.

C and **D** are incorrect. The titration is a weak acid/strong base titration, so the equivalence point would be at a pH greater than 7.

QUESTION 10 D

D is correct. Methyl orange is red at pH values less than 3.1. For methyl orange to change to red in an acid that has a low concentration of 0.01 M, the acid would need to be fairly strong. As hydrochloric acid is a strong acid, a 0.01 M solution would make the indicator change to red.

A is incorrect. Ethanoic acid is a weak acid, so a 0.01 M solution would not make the indicator change to red.

B is incorrect. Sodium chloride is a salt and does not affect pH.

C is incorrect. Sodium hydroxide is a strong base and would, therefore, have a very high pH.

SECTION 2

QUESTION 11 (11 marks)

a) **Titration 1:** curve B (*strong acid and strong base*)**Titration 2:** curve C (*weak acid and strong base*)**Titration 3:** curve A (*strong acid and weak base*)

[3 marks]

1 mark for each curve identified.

b) Any one of:

- Methyl red would be suitable for titration 3. Methyl red's pH range (4.4–6.2) falls fully within the vertical section of the titration curve (curve A).
- Bromocresol green would be suitable for titration 3. Bromocresol green's pH range (3.8–5.4) falls fully within the vertical section of the titration curve (curve A).
- Bromothymol blue would be suitable for titration 3. Bromothymol blue's pH range (6.0–7.6) falls fully within the vertical section of the titration curve (curve A).

[2 marks]

1 mark for identifying a suitable indicator.

1 mark for explaining why the indicator is suitable.

c) Any one of:

- Curve A indicates a buffer region. At the half equivalence point, there are equal concentrations of the base and conjugate acid. Ammonia dissociates according to the equation $\text{NH}_3(\text{aq}) + \text{H}^+(\text{aq}) \rightleftharpoons \text{NH}_4^+(\text{aq})$. This dissociation can shift to the right (using up the increased number of H^+ in an acidic solution) or to the left (creating more H^+ in a basic solution) to maintain a consistent pH.
- Curve C indicates a buffer region. At the half equivalence point, there are equal concentrations of the acid and conjugate base. Ethanoic acid dissociates according to the equation $\text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}^+(\text{aq})$. This dissociation can shift to the left (using up the increased number of protons (H^+) in an acidic solution) or to the right (creating more H^+ in a basic solution) to maintain a consistent pH.

[3 marks]

1 mark for identifying either curve A or C.

1 mark for stating that there are equal concentrations of acid and conjugate base at the half equivalence point.

1 mark for explaining that dissociation can shift right or left to maintain a consistent pH.

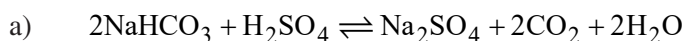
Note: Curve B does not have buffer region as a buffer region is not found on a strong acid/strong base titration curve.

d) At the half-equivalence point, the dissociation constant (K_a) is equal to the concentration of $[\text{H}_3\text{O}^+]$ because $[\text{HA}] = [\text{A}^-]$ (meaning that $[\text{HA}]$ and $[\text{A}^-]$ cancel each other out in the equation

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}) . \text{ Therefore, the pH is equal to } \text{p}K_a, \text{ so } K_a \text{ is } 10^{-\text{pH}}.$$

[3 marks]

1 mark for explaining that K_a is equal to $[\text{H}_3\text{O}^+]$.1 mark for stating that pH is equal to $\text{p}K_a$.1 mark for stating that K_a can be determined using $K_a = 10^{-\text{pH}}$.

QUESTION 12 (7 marks)

[2 marks]

1 mark for using the correct chemical formulas.

1 mark for balancing the equation.

b) $C(\text{H}_2\text{SO}_4) = \frac{n}{V}$

$$\begin{aligned} n &= C \times V \\ &= 0.0500 \times 0.00735 \\ &= 3.675 \times 10^{-4} \end{aligned}$$

ratio of $\text{NaHCO}_3 : \text{H}_2\text{SO}_4$ is 2 : 1

$$\begin{aligned} n(\text{NaHCO}_3) \text{ in } 10 \text{ mL} &= 2 \times (3.675 \times 10^{-4}) \\ &= 7.35 \times 10^{-4} \end{aligned}$$

$$\begin{aligned} \therefore n(\text{NaHCO}_3) \text{ in } 50 \text{ mL} &= 5 \times (7.35 \times 10^{-4}) \\ &= 3.675 \times 10^{-3} \end{aligned}$$

$$\begin{aligned} m(\text{NaHCO}_3) &= n \times M \\ &= (3.675 \times 10^{-3}) \times 84.01 \\ &= 0.3087 \text{ g} \end{aligned}$$

$$\begin{aligned} \text{percentage of NaHCO}_3 \text{ in baking powder sample} &= \left(\frac{0.3087}{2.00} \right) \times 100 \\ &= 15.4\% \end{aligned}$$

[5 marks]

1 mark for calculating the amount of H_2SO_4 .1 mark for calculating the amount of NaHCO_3 in 10 mL of the solution.1 mark for calculating the amount of NaHCO_3 in 50 mL of the solution.1 mark for calculating the mass of NaHCO_3 in the sample.1 mark for determining the percentage of NaHCO_3 in the sample.**QUESTION 13 (4 marks)**

[1 mark]

1 mark for identifying CN^- .

b) At 25°C , $K_w = K_a \times K_b$.

$$1 \times 10^{-14} = 6.2 \times 10^{-10} \times K_b$$

$$K_b = 1.61 \times 10^{-5}$$

$$\begin{aligned} \text{p}K_b &= -\log_{10} K_b \\ &= 4.8 \end{aligned}$$

[3 marks]

1 mark for using the formula K_w .1 mark for calculating K_b .1 mark for calculating $\text{p}K_b$.

QUESTION 14 (2 marks)

Acid X is the stronger acid. Acid X has a lower pK_a than acid Y. A low pK_a indicates a high K_a and greater dissociation.

[2 marks]

1 mark for identifying that acid X is the stronger acid.

1 mark for explaining that acid X's lower pK_a value indicates a high K_a and greater dissociation.

QUESTION 15 (3 marks)

$$K_a = 10^{-4.76}$$

$$= 1.7378 \times 10^{-5}$$

$$K_a = \frac{[H^+]^2}{[HA]}$$

$$[H^+] = \sqrt{K_a \times [HA]}$$

$$= \sqrt{(1.7378 \times 10^{-5}) \times 0.20}$$

$$= 0.0019 \text{ M}$$

$$\text{pH} = -\log[H^+]$$

$$= -\log 0.0019$$

$$= 2.73$$

[3 marks]

1 mark for calculating the K_a .

1 mark for calculating the concentration of H^+ .

1 mark for calculating the pH.

QUESTION 16 (3 marks)

During physical exercise, the increase in the amount of carbon dioxide (CO_2) would cause equation 1 to shift to the right, which would increase the amount of hydrogen carbonate ions (HCO_3^-) and hydrogen ions (H^+) produced. The increase in H^+ would then cause equation 2 to shift to the right in order to use up the excess H^+ , maintaining the pH at a consistent level.

[3 marks]

1 mark for explaining that both equations would shift to the right.

1 mark for explaining that the shift in equation 1 would result in more HCO_3^- and H^+ .

1 mark for explaining that the shift in equation 2 is in response to the increase in H^+ and maintains the pH by using up the excess H^+ .